

SNEHA PANDIT

LinkedIn: www.linkedin.com/in/sneha-pandit-19010729b

Github: github.com/snehapandit2006

Email: sneha20061901@gmail.com

Mobile: +91-7881184884

EDUCATION

- **GLA University** | B.Tech in Computer Science & Engineering (Spec. AI & ML) | **Expected: 2027**

TECHNICAL SKILLS

- **AI Systems & LLM Engineering:** Retrieval-Augmented Generation (**RAG**), **LangChain**, LangGraph (Stateful Agent Workflow Concepts), **Vector Databases (ChromaDB, Pinecone)**, PyTorch, Prompt Engineering, Multi-Agent Workflows, Transformer Models, NLP, Hugging Face Transformers, RoBERTa, Scikit-learn, Pandas, NumPy.
- **Backend Systems & Databases:** Python, **FastAPI**, Node.js, Express, AsyncIO, BullMQ, Redis, Prisma ORM, SQL, RESTful APIs, WebSockets, MongoDB, SQLite.
- **Cloud & Infrastructure:** **Azure Certified**, Cloud Infrastructure & Deployment Concepts, **Docker**, Linux Environments, GitHub Actions, Git.
- **Additional Languages:** TypeScript, JavaScript, Java, C++.

AI & BACKEND ENGINEERING EXPERIENCE

Bhumi: Smart Land Governance AI Engine | Python, LangChain, FastAPI, Vector Databases | [GitHub Repository](#) | [Technical Report](#)

- Architected an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline using **LangChain** to ingest, segment, and index dense, unstructured legal land records.
- Optimized similarity search routing using **Hugging Face embedding models** and **Vector Databases** to resolve domain-specific user requests with high accuracy.
- Engineered a high-throughput asynchronous backend layer using **FastAPI** to process concurrent document indexing streams and user queries.
- Recognized as a **National Finalist at the Smart India Hackathon (SIH) 2024** after competing in the 36-hour Grand Finale against thousands of teams nationwide.

Sentia: Multimodal Emotion Drift Analytics Platform | Python, PyTorch, FastAPI, WebSockets | [GitHub Repository](#) | [Technical Report](#)

- Built a parallel multimodal inference pipeline in Python using **ThreadPoolExecutor** to reduce **P95 latency below 4 seconds**.
- Fine-tuned a **RoBERTa-base transformer** on a normalized 7-class schema of the GoEmotions dataset to achieve a **macro F1-score above 0.70**.
- Implemented a confidence-weighted late-fusion architecture combining text embeddings with a facial expression CNN to improve multi-turn classification accuracy.
- Tracked emotional drift across multi-turn user sessions using **Total Variation Distance (TVD)** over 7-class probability distributions.
- Deployed and tested backend inference APIs in containerized Linux environments using **Docker**, simulating cloud deployment workflows.

Panopticon: Distributed Political-Economic Data Observability Engine | TypeScript, Node.js, Express, Prisma, BullMQ | [GitHub Repository](#) | [Technical Report](#)

- Engineered a high-throughput distributed tracing backend using **TypeScript and Prisma ORM** to map complex asset and economic data models.
- Developed asynchronous data-ingestion workers via **BullMQ** to process over **10,000 structured events/day** across distributed task queues.
- Integrated a modular **LangChain** processing layer into worker threads to parse unstructured media feeds and execute automated entity extraction.
- Secured backend endpoint routing against unauthorized administrative data injection using **JWT authentication** and cryptographic verification.
- Configured **Azure-based** deployment workflows and infrastructure concepts for scalable distributed AI backend services.

OPEN-SOURCE CONTRIBUTIONS & LEADERSHIP

Project Admin / Contributor | GirlScript Summer of Code (GSSoC) | 2025 – 2026

- **Project Admin (2026):** Moderated code health, structured detailed contribution guidelines, and reviewed/merged contributor Pull Requests across deep learning and backend repositories.
- **Contributor (2025-2026):** Implemented modular feature extensions and refined data processing scripts across open-source AI/ML codebases.
- **Repository Engineering:** Maintained public markdown specifications and automated **GitHub Actions CI/CD workflows** to guarantee main branch stability.

AWARDS & ACHIEVEMENTS

- **National Finalist** | Smart India Hackathon (SIH), Ministry of Education, Govt. of India | 2024
- **First Place Winner** | Smart India Hackathon Internal Nomination, GLA University | 2024
- **National Phase 2 Selection** | IBM Pr आरंभ Hack Challenge | 2025